Democratization theory

A game-theoretic approach

Lecture 08

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Table of contents

1. Introduction

2. Democratization model

3. Repression as an option

4. Coups and democracy stability

Last lecture:

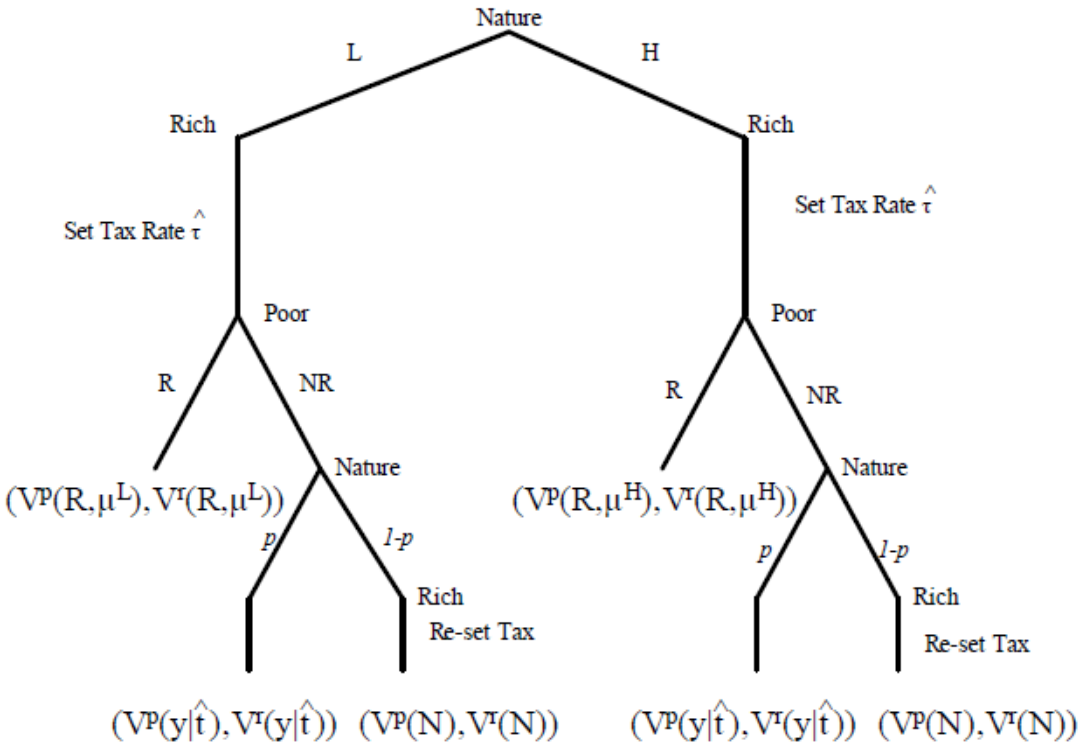

Figure: Game of Promises

- Two states representing de facto power: $S = L$ or H
- Poor people payoff (revolution): $V^p(R, \mu^S) = \frac{(1-\mu^S)\bar{y}}{1-\delta}$, $\mu^L = 1$
- Probability p captures commitment problem
- $p = 1$: full commitment
- $p = 0$: complete inability to commit
- Rich people payoff (revolution): $V^r(R, \mu^S) = 0$

- Focus: $S = H$
- If $V^p(R, \mu^H) > V^p(N)$, revolution threat is real ($\theta > \mu^H$)
- What tax the elite will offer to dissipate the threat?
- $V^p(N, \tau = \hat{\tau}) \geq V^p(R, \mu^H)$. This means:

$$y^p + p(\hat{\tau}(\bar{y} - y^p) - C(\hat{\tau})\bar{y}) \geq \frac{(1 - \mu^H)\bar{y}}{1 - \delta}$$

or

$$\mu^H \geq \theta - p(\hat{\tau}(\theta - \delta) - (1 - \delta)C(\hat{\tau}))$$

- Maximum commitment is $\hat{\tau} = \tau^p$
- Defining $\mu^* = \theta - p(\tau^p(\theta - \delta) - (1 - \delta)C(\tau^p))$

Proposition: There is a unique subgame perfect equilibrium in the game described before, and it is such that:

- If $\theta \leq \mu^H$, $\tau^N = 0$ and poor **do not** revolt.
- If $\theta > \mu^H$, constraint is binding in the high state. In this case:
  - If $\mu < \mu^*$, promises by the elites are not credible to avoid revolution.
  - If $\mu \geq \mu^*$, promises are credible and can avoid revolution.

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- What about the role of the middle class?

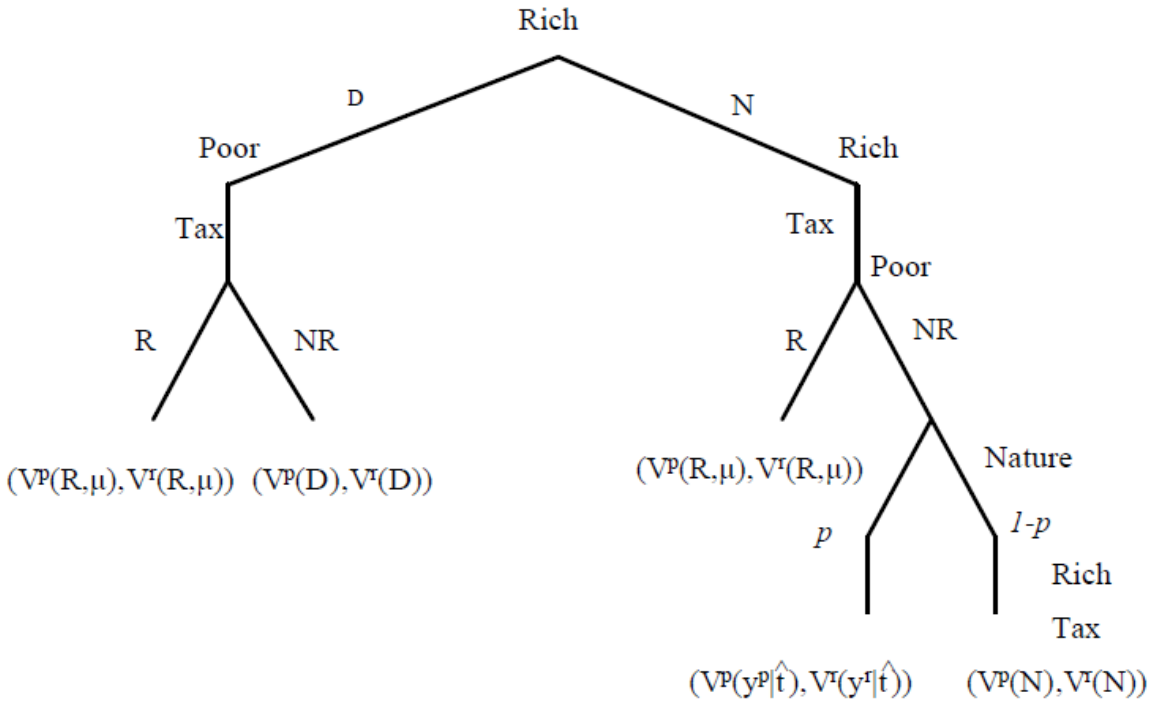

Figure: Democratization game

Proposition: There is a unique subgame perfect equilibrium in the game described before, and it is such that:

- If $\theta \leq \mu$, then the revolution constraint does not bind and the elites can stay in power without democratizing or redistributing income.
- If $\theta > \mu$, revolution constraint is binding. In this case:
  - If $\mu \geq \mu^*$, elite promises are credible and can avoid revolution setting tax $\hat{\tau}$.
  - If $\mu < \mu^*$, promises by the elites are not credible to avoid revolution. Then if $V^p(D) \geq V^p(R, \mu)$, elites democratize. If $V^p(D) < V^p(R, \mu)$, there is a revolution.

- Repression as an option to the elites: similar to revolution
- κ: represents a cost parameter to repress
- Payoffs: $V^P(O|\kappa) = (1-\kappa)y^p$ and $V^r(O|\kappa) = (1-\kappa)y^r$
- Rest of the three is equal to the previous game

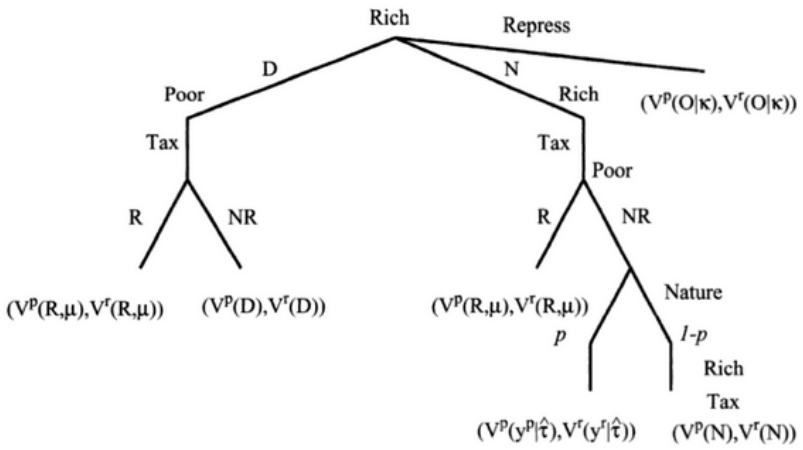

Figure: Democratization game with repression as option

Define:

- $\hat{\kappa}$ by $V^r(O|\hat{\kappa}) = V^r(N, \hat{\tau})$ or $\hat{\kappa} = \frac{p}{\theta}\left(\delta C(\hat{\tau}) - \hat{\tau}(\delta - \theta)\right)$
- $\tilde{\kappa}$ by $V^r(O|\tilde{\kappa}) = V^r(D)$ or $\tilde{\kappa} = \frac{1}{\theta}\left(\delta C(\tau^p) - \tau^p(\delta - \theta)\right)$
- Note that $\hat{\kappa} \leq \tilde{\kappa}$

Proposition: There is a unique subgame perfect equilibrium in the game described before, and it is such that:

- If $\theta \leq \mu$, then the revolution constraint does not bind and the elites can stay in power without repressing, democratizing or redistributing income.
- If $\theta > \mu$, revolution constraint is binding. In this case:
  - If $\mu \geq \mu^*$ and $\kappa > \hat{\kappa}$, repression is relatively costly and the elites redistribute income to avoid a revolution.
  - If $\mu < \mu^*$ and $\kappa < \tilde{\kappa}$, or $\kappa \geq \hat{\kappa}$ and $V^p(D) < V^p(R, \mu)$, or if $\mu \geq \mu^*$ and $\kappa < \hat{\kappa}$, then the elites use repression.
  - If $\mu < \mu^*$, $V^p(D) \geq V^p(R, \mu)$ and $\kappa \geq \tilde{\kappa}$, concessions are insufficient to avoid revolution and repression is relatively costly so the elites democratize.

- After a democracy takes place, elites have a coup option
- φ: represents a cost parameter to coup (similar to revolution parameter)
- Payoffs: $V^P(C, \varphi) = (1 - \varphi)y^p$ and $V^r(C, \varphi) = (1 - \varphi)y^r$
- Options: poor set tax. Rich people decide to coup or not

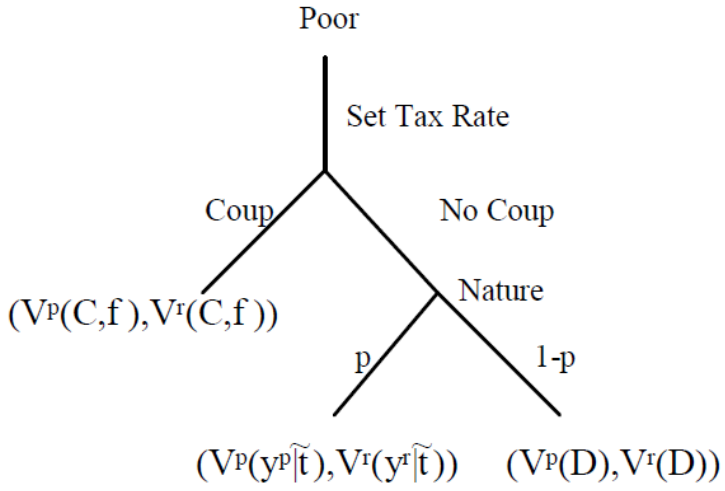

Figure: Coup game

- Coup constraint: $V^r(C, \varphi) > V^r(D)$ implies $\varphi < \frac{1}{\theta}(\delta C(\tau^p) - \tau^p(\delta - \theta)) := \hat{\varphi}$
- Promisse to tax less: poor people have the option of setting a lower tax
- $V^r(C, \varphi) \leq V^r(D, \tau^D = \tilde{\tau})$
- Most attractive tax to rich people: $V^r(D, \tau^D = 0) = V^r(C, \varphi^*)$, where φ^* will be a threshold
- Therefore, $\varphi^* := \frac{1-p}{\theta}(\delta C(\tau^p) - \tau^p(\delta - \theta))$

Proposition: There is a unique subgame perfect equilibrium in the game described before, and it is such that:

- If $\varphi \geq \hat{\varphi}$, then democracy is consolidated (cost is to high to mount a coup).
- If $\varphi^* \leq \varphi < \hat{\varphi}$, democracy is semiconsolidated. The citizens set a lower tax then the desirable such that $V^r(D, \tau^D = \tilde{\tau}) = V^r(C, \varphi)$
- If $\varphi < \varphi^*$, then a coup is inevitable.